

DNSFormer: A Multi-Objective Transformer Model for Stealthy DNS Attack Detection

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Abstract- The Remote Access Trojans and the data exfiltration represent some of the persistent and hidden threats in modern cybersecurity as, the Domain Name System has become a major threat particularly affecting corporate, government and high value infrastructure. Threats such as DNS amplification and DNS Fast-flux play a major role in exploiting DNS for remote access trojans and data exfiltration through tunnelling, which is one of the stealthiest methods as it hides the command-and-control communication within legitimate DNS traffic, making detection challenging as it is bypassing firewalls and IDS. The traditional IDS often show high false alarm and moreover they lack in balancing between efficient detection and system performance, which eventually not suitable for real-time deployment. To overcome limitations, this study proposes a DNSFORMER model for DNS classification and it enhance through multi-objective optimization with NSGA-2(Non-dominated Sorting Genetic Algorithm II). NLP-based packet sequence modelling, statistical analysis, and reinforcement learning are employed to identify and block unauthorized data transfer channels. The optimization specifically targets 12 objectives: seven classification metrics (Accuracy, Precision, Recall, F1-score, True Positive Rate, False Positive Rate, and False Negative Rate) and five Quality of Service (QoS) metrics (Response Time, Latency, Throughput, Reliability, and Availability).The DNS amplification and the fast-flux are being captured In separate algorithms approaches, as how IP rotation occurs and they malform as an victim's IP ,in order to cause Denial of Service (availability attack) and Stealthy hosting & persistence (confidentiality + integrity attack). Comparative analysis shows how an NSGA-2 and other algorithms perform on that 12 objective. By incorporating layered security intelligence, this research presents a robust, scalable, and future-ready Defence mechanism against DNS-based RAT infiltration and hidden data exfiltration. There by strengthening cybersecurity resilience in mission-critical environments.

Keywords - Remote Access Trojan(RATs),Data exfiltration, Quality of Service, Fast Flux, DNS Amplification, DNS Former model.

Introduction

In recent years, the dependence on the digital market has grown significantly, which alternatively showed a rapid evolution of cyber threats. Among the wide variety of cyber threats, Remote Access Trojans (RATs) and Data Exfiltration are major concern because of their stealthy and persistent in nature. The Domain Name System (DNS) plays a major role in network communication and is often exploited by attackers using methods like DNS spoofing, DNS tunneling and DNS Amplification attacks.

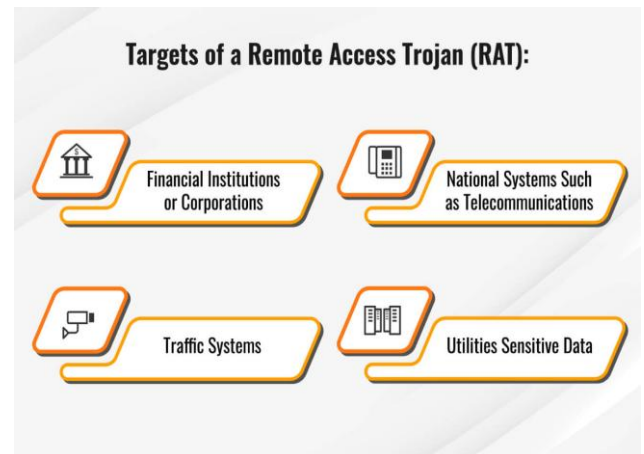


Figure 1 Targets of RAT

RATs are malware that provide attackers with full remote control over the victim's system. By injecting malicious communication within legitimate DNS queries and responses, attackers can easily bypass the conventional firewalls and intrusion detection systems (IDS) and then data exfiltration occurs, it involves the unauthorized transfer of sensitive information from a system or network to an external destination controlled by an attacker. Since most firewalls, proxies, and IDS allow DNS traffic by default, DNS-based attacks are difficult to detect and can lead to data theft, credential harvesting, and further malware deployment. Traditional defences such as backups and recovery methods are not sufficient to protect against such attacks.

Fast-flux is a technique that rapidly using in DNS records, particularly IP addresses, as it will be majorly associated with malicious domain names. This makes it hard to track and shut

down the malicious domain, as blocked IP addresses are quickly replaced. There are two types of fast-flux, the first one is single-flux, which rapidly changes A records, and double-flux, which frequently changes both A records and authoritative name server records. The traditional Detection is more difficult due to rapid IP rotation and when we overlap between benign Content Delivery Networks (CDNs), enabling attackers to maintain resilient malicious infrastructures.

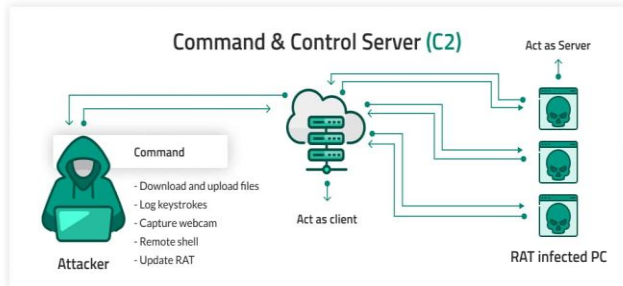


Figure 2 Command & Control Server (C2)

DNS amplification is a type of Distributed Denial of Service (DDoS) attack that exploits DNS resolver, as in this the attacker sends a spoofed DNS queriers to public servers using the victim's IP causing the servers to replay with the amplified responses to the victim, then that will respond with a much longer DNS data packets to the victim resulting overwhelming the target. Attackers send small spoofed DNS queries spoofs victim's source IP making it look like victims requesting it, then the open DNS resolver replies to it with a large amplified response as it thinks it came from an legitimate user as an result the victim's get flooded with massive traffic , and here the major concern is the small query can generate huge responses as these responses are being directed towards the spoofed Victim's IP address, which is one of the main reason the attacker prefer to use the DNS amplification as it cause major damage to the victims on the other side it only cause a little bandwidth consumption for the attacker. This results in massive traffic flooding the target while requiring minimal effort and bandwidth from the attacker. These attacks commonly target sectors such as financial services, government organizations, healthcare, cloud providers, and ISPs. Mitigation techniques include rate limiting, ingress filtering, securing DNS resolvers, using CDNs, and deploying anomaly detection systems, though complete prevention remains challenging.

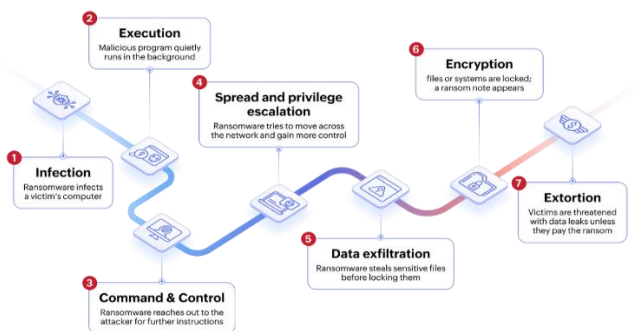


Figure 3 The Flow of RAT Attack

Although many studies have done research on DNS-based attacks, the most common way of addressing the issues was by signature-based or offline analysis, which use to be

inefficient to detect and analyse different attack behaviours in real time. Moreover, many solutions only consider accuracy without considering real-time constraints such as latency, scalability and system performance, while this enables it and focus on detecting RAT communication, fast-flux behaviour, and DNS amplification in modern environments.

Related Works

Ransomware has become a major cybersecurity threat as it is used to exfiltrate, encrypts and also used to delete user data or change the data causing many serious problems to victims. The traditional defences such as backups and recovery methods are not sufficient. Hence, the future research must focus on data exfiltration collaboration with the industry[1]. Data breach attacks are unique, as attackers exfiltrate data from the targeted systems. The dataset redefined with ten features, and balanced using (SMOTE+ENN) technique. They benchmarked three resampling methods, five feature selection techniques and five ML algorithms as their findings from real-world study say they achieve higher accuracy rather than other algorithms[2].

The studies on the DNS- based data-exfiltration was majorly dependent on offline analysis, as this allowed many attacker to retrieve a large amount of data and also without being detected. To address this issue, the study have proposed an information- based Heavy hitters(ibHH), which says that how much information is sent to registered domains, and their evaluation shows that it can process high volumes of queries with a very low false positive rates[3]. Data exfiltration is a serious cybersecurity threat in edge computing environments. When a large amount of data are being transferred and processes and moreover it a sensitive information. This study shows that this can improve detection performance and make real-time application more efficient by using adaptable techniques along with AI [4].

QUIC protocol are widely used in many multiplexing features and additionally it is secure, efficient and resistant to many censorship as that is mainly encrypted and a little different form TCP and UDP, since it is encrypted it is hard to analyze the QUIC connection, which makes it challenging to identify normal and malicious connection[5].To counter the data exfiltration ransomware an advanced decoy file is generated as this are placed in entire networks to attract the attacks or ransomware activities and it is done using Generative Adversarial Networks (GANs), and this works in real-time and enables fast response. [6].

The existing benchmarks in large language model are generally used for attacks, and they are not focusing on data-exfiltration, this recent works have indicated that the prompt injection can leak that sensitive information when the prompt can do force tool- calling, and these are tested with AgentDojo benchmark[7].Reverse Firewall provides a defence mechanism for cryptographic protocols against subversion attacks, as in the prior studies established robust guarantees a practical implementation of OP-TEE to validate further security and provides a better real-time applications for any cryptographic functionality of reverse firewall[8].

Battery powered embedded systems (BES) are integrated into environments more, as this are causing more security

problems such as internal vulnerabilities, in real-world applications, which are similar to E-trojans. This research can detect many attacks and enables to highlight a better security mechanisms[9]. This study is to analyse the silly putty malware by combining the static and dynamic techniques, using sandbox and Virtual Machines. The static helps in identifying the characteristics of malware and the dynamic analyse and provide the behaviour of the malware in the isolated environment, as this improves the observing and analysing the malware in real-time[10].

This study reveals that developing algorithms to detect and analyze the Remote Access Trojans(RATs) and due to their stealthy nature and more in number of attacks in occurrence, many model were used such as VGG-16, VGG-19, and ResNet50, to test the effectiveness of the RAT[11]. This study uses the Mega RAT version 1.5 Beta, they deployed an experimental setup with multiple virtual machines to simulate attacks and figure out either static or dynamic is more efficient to check that, it was done on victim on a specific network,12].

This study shows that the malicious modification to the printed circuit board (PCB) are called as hardware Trojans, as this attack are possible only, when the other third parties alter the PCBs pre-manufacturing or post-manufacturing mainly in security related application such as industrial control systems [13].One of the most damaging attacks in cyber security is Advanced Persistent Threat (APT) as it has an adaptive behaviour and additionally the existing methods face many difficulties with data integrity issues and low real-time performance. To Address these challenges, this research has introduced APTShield, which is much efficient in detecting real-time environment for Linux system and it also detects many vulnerabilities with low false positive rate[14].

Through remote access the FPGAs(Field programmable gate arrays) offer runtime reconfiguration, which has an pros of utilization of the resource and as for the cons this can also enable or introduce high risk security vulnerabilities, as this could insert the hardware trojans(HT) in configuration bitstream, bypassing any security, which is used for protection and verifications measures. To address those limitations a supervised learning using deep recurrent neural networks (RNN) was developed, and has two RNN architectures: basic RNN and long short-term memory (LSTM) [15].

The reliance on hardware technologies has caused vulnerabilities rise, as the technologies increases, the cloning the devices have been increased a lot. To address this issue, the research have proposed a physically unclonable functions (PUFs), and the techniques to detect hardware trojans in circuits and for authentication. Moreover, this have been introduced in order to defense against side-channel attacks, which is the major cause for the information leak through power usage or electromagnetic signals.16].The increase in use of deep neural networks as for decision making, by knowing their major role in real-time application, recent studies a have showed how to exploit it. To address this issue, trace the traditional trojans to NN Trojans, the study had provided a various attacks and defense strategies[17].

Materials and Methods

The Figure 4 is a fast flux. It is an DNS-based technique in which malicious domains rapidly change their IP addresses, as it hard to track and shut down the malicious domain, This technique hides the true location of the command-and-control(C2) infrastructure, by doing so this allows attackers to host services like botnets, phishing sites, and malware distribution servers on the network of compromised hosts, that act as proxies.

Fast-flux are operated in two forms: single-flux and double-flux, the single flux is the domain is associated with large and rapidly rotating of IP addresses and the double flux is that both the IP addresses as well as authoritative nameservers are also being rapidly changed, the DNS server responds with rapidly rotating Ips with short TTL(Time to Live). By hiding the real C2 servers it evades many detection methods and any compromised machine can act as a proxy node, which shows scalability.

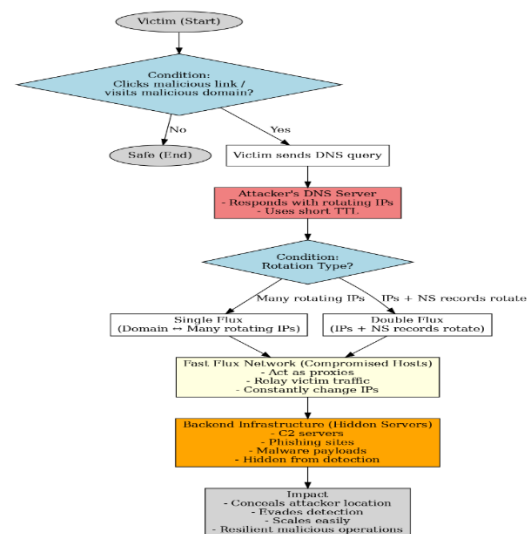


Figure 4 Fast Flux Flow Chart

The Figure 5 Shows the DNS amplification attack flow. It is a type of Distributed Denial of service (DDoS) attack, which affects the victim by spoofing the victim's IP, that exploits open DNS resolver to send a high-volume traffic towards victim. The attacker performs this attack by sending malicious queries to servers while spoofing the victim's IP address as a source, then the resolver responds with a much larger DNS packet to victim results in overwhelming the target.

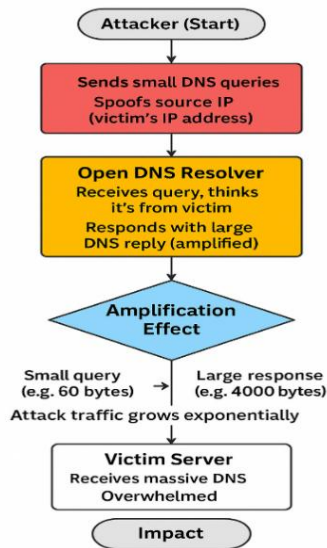


Figure 5 Dns Amplification Flow Chart

Result Analysis

The main objective of this study is to show that the DNS - based attacks such as RAT communication, Data Exfiltration, Data amplification and Fast-flux attack are being effectively detected using a Transformer-based DNSFormer, while following real-time operational constraints, and using the proposed DNSFormer model is evaluated by different datasets, and it's performance using 12 objectives using NSGA-II and moreover, a comparative analysis of QoS across optimization algorithms and pareto-based trade-off for real-time system performance. This study is conducted to show the high detection performance and real-time system performance by optimizing DNS-based detection framework and Quality of service requirements.

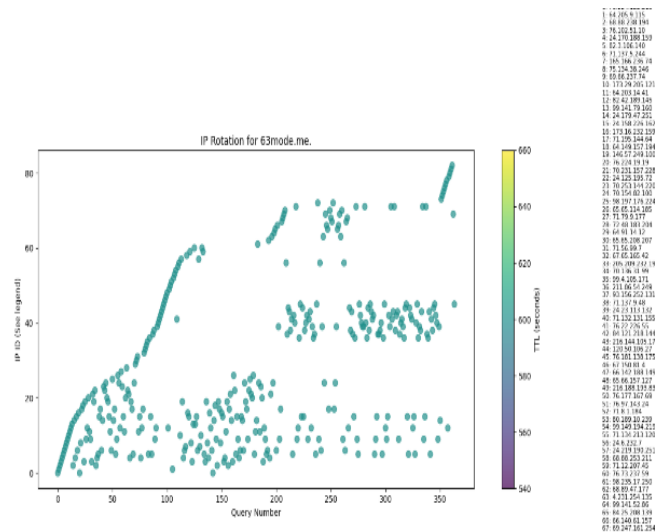


Figure 6 Fast Flux Analysis

The Figure 6 shows the 62mode.me. It is an suspicious hostname that the analysis is tracking and showing us how the rotation is DNS responses among the dozens of Ip's as this is the fast flux behaviour, as the each point in the X-axis represent one DNS query made for the domain 63mode.me the values keep on goes from 1 to many, this indicates that many DNS Lookups were performed and TTL (Time-to-Live) is a field, where the DNS record tells how long (in seconds) a resolver can detect and cache the IP address before asking the DNS server again, as the IP list on the right links

the plotted IDs to actual resolved IPs (e.g., 68.71.130.164, 178.74.17.137, etc.).

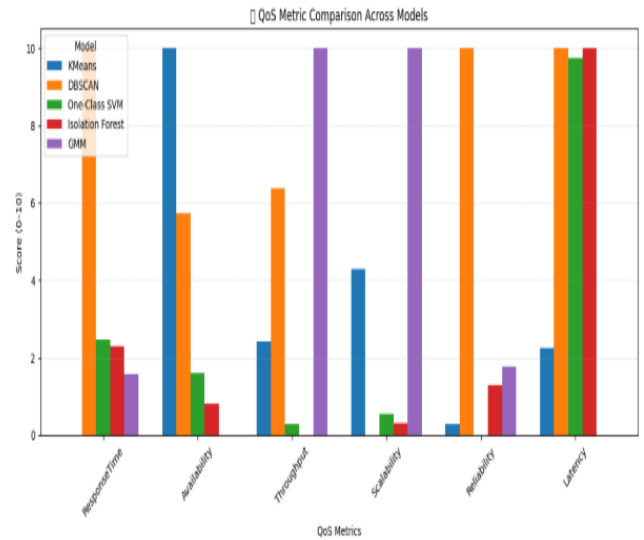


Figure 7 Qos Metric Comparison Access models

This figure 7 shows a five different algorithm as each and every metrics there are only four bars as this indicate that, the missing one bar's score is 0, and the results indicate the varying strengths among the algorithms.

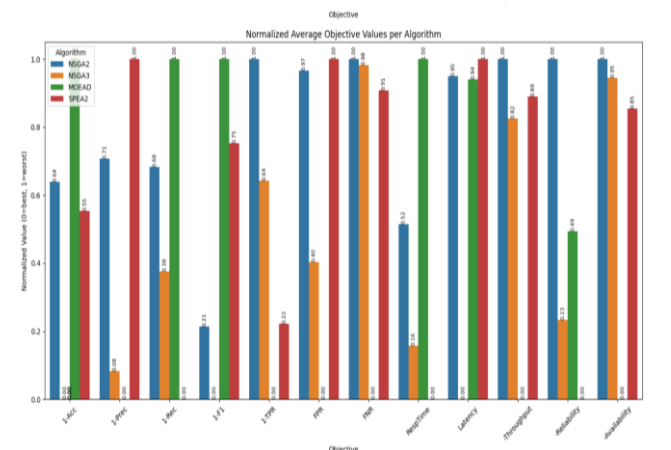


Figure 8 Comparative Multi- Objective algorithms

The Figure 8 shows the normalized Average Objective values per Algorithm is being calculated across four different algorithms in a normalized score of (0-1), by considering the 12 objective, this study proposed a multi objective optimised model using NSGA-II and the model which is very close to zero indicates the best algorithm.

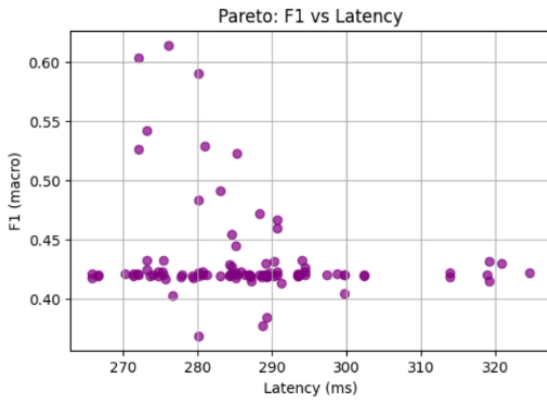


Figure 9 Pareto graph of NSGA-II(F1 vs latency)

This Figure 9 shows that it is a pareto graph that shows F1 vs Latency and the latency(lower=better) and the F1-score (higher = better) and each purple dot shows, one solution from the NSGA-II and many solutions have high latency and have low F1-score, if needed a better F1-score, the solution should be able to tolerate the higher Latency.

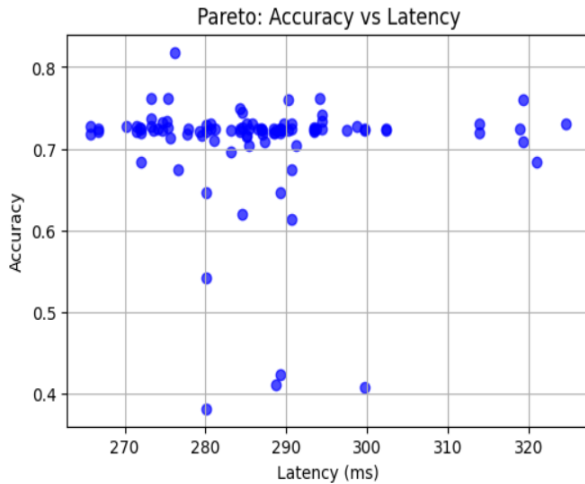


Figure 10 Pareto graph of NSGA-II(Accuracy vs Latency)

The Figure 10 is a pareto graph that shows Accuracy vs latency (the lower = better) and each blue dot shows an one solution from the NSGA-II, as some solutions trade off's are very low as the accuracy and the latency are inversely related as they can't be optimized fully.

Datasets

To evaluate the proposed DNSFormer framework, multiple datasets have been used to analyse different attack behaviours and system performance characteristics, as this includes DNS traffic, fast-flux data, DNS amplification data, and a Quality of Service(QoS) dataset. The DNS dataset consists of both benign and malicious query DNS-response, which is collected from real world and controlled environments, and the features used for evaluation are query length, query type, time-to-live(TTL), response size.

The fast-flux dataset contains the DNS records, which are collected from fast flux attack of rapid IP rotation, and the features used for evaluation of fast flux are IP diversity,

domain entropy, frequency of IP changes, name server, as this dataset helps to detect the command and control(C2) used by malware and botnets. The DNS amplification dataset contains the attack traffic generated through DNS resolver, as it captures components such as abnormal traffic rates, query to response size, this dataset helps to detect abnormal response size and DOS behaviour.

Although containing different attack detection datasets, a QOS based dataset is introduced to access real time system performance, as this includes latency, response time, throughput, accuracy, availability, and reliability under various traffic. This ensures the proposed system maintains efficiency while detecting attacks, moreover all datasets are pre-processed and normalized, resulting in detecting attacks along with Quality of Service constraints with real-world environments.

Conclusion and Future Work

This study proposes a DNSFormer, a Transformer-based DNS classification architecture as this study took it further with performance evaluation to detect the stealthy Remote Access trojans(RATs), DNS data exfiltration and used different attacks such as DNS Amplification and Fast-Flux, showcase how the Domain Nam System (DNS) is being targeted and exploited. We have formulated a 12 Objective optimization (seven classification metrics + five quality of service) using NSGA-II and also evaluated it with other comparative algorithms(NSGA-III,MOEA/D, and SPEA2) for each metrics showing meaningful trade-offs between detection quality and operational and additionally included the fast-flux and DNS Amplification detection methods and an DNS former model similarly to detect the DNS tunnelling and DNS based attacks as this provides an scalable ,real-time capable defence against these stealthy attacks .

We can extend the future work of detecting the DNS over HTTPS (DOH), DNS over TLS, QUIC as these all hide the malicious traffic increasingly in encrypted channels , we can also fuse the DNS formal model with threat feeds, SIEM/SOAR to detection purpose and automated incident purpose and further instead of NSGA-II we can implement the NSGA-III and we should validate the DNSFormer in live enterprise under high traffic loads and to test on larger datasets of real-world datasets which consists of more critical infrastructure across every different conditions.

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